FOOD AND AGRICULTURE SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military-Action Extension Methodology
Function-Primary Facility-Class Architecture — the Distributed Web Sector

DBA-FA: Food and Agriculture Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

Version 0.5 — DRAFT

October 2026

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For review by engineers, regulators, producers and processors, and food-safety and policy analysts responsible for the resilience and integrity of the food and agriculture supply.

SUGGESTED CITATION

T. Roxey, "DBA-FA_Sector_Framework — Food and Agriculture Sector Framework," Version 0.5, in CIKR Design Basis Examples, Edition 1 (November 2026), Eclectic Technologies, November 2026. https://doi.org/10.5281/zenodo.23863986

This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Food and Agriculture sector. It is the parent instrument for the DBA-FA series and applies the DB → DBA → DBA-MA three-layer analytical progression to the functions on which the food supply depends: agricultural production, food processing, and storage and distribution, together with the food-safety-and-biosecurity function that governs them all.

1.2 What This Document Does Not Do

This document does not supersede food-safety regulation or agricultural policy. It does not restate the Bright Line or Command Broker, inherited by reference from FD-BL. It does not re-scope the production of agricultural inputs: fertilizer and pesticide manufacturing is governed by the Chemical sector and veterinary pharmaceutical manufacturing by the healthcare sector, and both are inherited here as boundaries rather than duplicated.

1.3 Function-Primary Organizing Principle

Placement follows the consequence chain. The sector is organized by function — the production of food, its processing, and its storage and distribution — and each function is placed by the food-safety or food-supply consequence its failure produces, sub-indexed by commodity type and perishability.

1.4 Series Placement — DBA-FA as a Standalone Sector Series

DBA-FA is a standalone sector series, a peer of the energy, water, chemical, communications, transportation, financial, healthcare, critical-manufacturing, emergency-services, defense-industrial-base, and nuclear series. It inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination.

1.5 A Dual-Consequence, Distributed Web Sector

Food and Agriculture has two features that shape its design basis. The first is a dual consequence. Like most sectors it carries a supply consequence — the availability of food to the population — but it also carries a safety consequence, and the two are governed differently. The safety consequence is the water sector's contamination model applied to food: a deliberate contamination of the food supply produces public-health consequences that are geographically broad, temporally delayed from the point of introduction, and difficult to attribute rapidly. The sector taxonomy names the parallel exposure directly in its vulnerability to disease outbreaks in crops and livestock. The framework therefore inherits the contamination-consequence structure the water sector developed and adds the biosecurity dimension of animal and plant disease.

The second feature is that Food and Agriculture is the most distributed and the most cross-sector-coupled sector in the framework. Its production is geographically dispersed across a vast number of farms, so — unlike a hub airport or a large power transformer — no single production site is a critical facility; the design basis concentrates instead on the chokepoints, at food processing, where a small number of large plants handle national volume, and at the input and cold-chain nodes. And the sector depends, as the sector taxonomy records, on five other sectors at once: energy for irrigation, processing, and refrigeration; water for irrigation and processing; transportation for farm-to-market and distribution; chemical for fertilizers and pesticides; and financial services for the commodity markets and crop insurance that underwrite it. It is the sector where the cross-sector web is densest, and its design basis is written around that web rather than around a set of isolable facilities.

Section 2 — The Food and Agriculture Sector and Its Regulatory Context

2.1 A Split-Jurisdiction, Consensus-Governed Sector

Food and Agriculture is governed under a split federal jurisdiction. The Department of Agriculture regulates agricultural production and the safety of meat, poultry, and processed-egg products; the Food and Drug Administration regulates the safety of most other foods and of animal drugs, and administers the Food Safety Modernization Act; the Environmental Protection Agency regulates pesticides and the water quality that agriculture affects; and state departments of agriculture carry substantial operating authority. The Department of Agriculture and the Department of Health and Human Services share the Sector-Specific Agency role.

2.2 Regulator-Per-Function Inheritance

The applicable authority varies by function and commodity: agricultural production inherits USDA and state authority; food processing inherits the USDA meat-and-poultry inspection regime and the FDA Food Safety Modernization Act framework; storage and distribution inherit the FDA and the cold-chain and transport regimes shared with the transportation sector; and the food-safety-and-biosecurity function inherits the inspection and disease-surveillance apparatus that spans USDA and FDA.

2.3 The Regulatory Gap

The sector carries extensive food-safety regulation, but no instrument states a design basis in consequence terms — the set of contamination and supply-disruption events the food system must be demonstrated to withstand, expressed as safety and supply-continuity requirements. Food-safety compliance is measured in inspection and process controls; a design basis measured in demonstrated withstand capability against deliberate contamination and supply denial does not exist.

Section 3 — Function-Primary Taxonomy

3.1 The Function Classes and the Commodity / Perishability Sub-Index

The sector is decomposed into three function classes — agricultural production, food processing and manufacturing, and storage, distribution and retail — and one coordination class, the food-safety-and-biosecurity coordination layer. Each is sub-indexed by commodity type and by perishability, so that the design basis load a given facility carries reflects both what it handles and how time-critical its cold chain is.

3.2 Agricultural Production (AG)

The production function covers crop production, livestock operations, dairy farms, and aquaculture. Its governing consequences are the biosecurity consequence — a crop or livestock disease outbreak that removes production at scale — and the input-contamination consequence. Its defining property is distribution: no single farm is a critical facility, so the design basis addresses the sector-scale disease and contamination events rather than site loss.

3.3 Food Processing & Manufacturing (FP)

The processing function covers meat and poultry processing, dairy processing, grain milling, beverage manufacturing, and packaged-food production. It is the sector's concentration chokepoint: a small number of large processing plants handle a national share of throughput for their commodity, so a processing plant is the point at which the distributed production system becomes a concentrated, high-consequence facility — and the point at which a deliberate contamination enters the food supply at scale.

3.4 Storage, Distribution & Retail (DR)

The distribution function covers grain elevators, cold-storage facilities, food warehouses, distribution centers, and the retail grocery and food-service tier. Its governing consequence is loss of the farm-to-table logistics on which a just-in-time food system depends, and its defining dependency is the cold chain, which it shares as a boundary with the transportation sector and which the perishability sub-index tracks.

3.5 OpsCo — Food Safety & Biosecurity (coordination class)

The coordination function is carried by a coordination class spanning the USDA and FDA inspection systems, the Food Safety Modernization Act framework, and the animal- and plant-disease surveillance apparatus. It is the sector's detection-and-response layer — the function whose degradation lets a contamination or outbreak spread undetected — and the locus at which the AI-Governance Design Basis is implemented: the Bright Line placement and Command-Broker obligation for machine-learning and generative-AI systems in precision agriculture, automated feeding and processing control, and food-safety analytics.

† The AG / FP / DR and OpsCo designators were ratified 2026-08-09.

3.6 The Dual Consequence Metric

The sector requires two consequence measures. On the supply axis it adopts a food-supply-continuity duration — the interval a served population can be sustained from stored and in-transit supply before shortage, tracked against the just-in-time character of the modern food system. On the safety axis it adopts the water sector's contamination-consequence structure — geographically broad, temporally delayed, attribution-resistant — measured by exposed population rather than by duration. As in the defense sector, the two axes must be stated explicitly and not collapsed into one. *Food Processing and Manufacturing* (DBA-FA-FP-FC2) calibrates the two measures as five parameters, supply share, supply-continuity duration, exposed population, time to detect and hold, and time to restart, and keeps the two axes separate as this section requires (*Food Processing and Manufacturing* §3.3, §10.2); this framework takes the calibration up at its next conformance pass, after the agricultural-production and distribution classes have been read against it.

Section 4 — The Food and Agriculture Sector Design Basis

[Skeleton — to be developed.] The design envelope will be organized by event category, spanning the supply and the safety axes, with a consequence-severity inclusion criterion consistent with the series.

4.1 Design Basis Principles

The criterion for inclusion is food-safety or food-supply consequence severity — potential for population-scale public-health harm from contamination, for a crop or livestock disease outbreak that removes production at scale, or for a supply disruption that produces shortage — not event likelihood. Candidate bounding events, to be developed: deliberate contamination introduced at a processing chokepoint; a crop or livestock disease outbreak; loss of a concentrated processing capability; cold-chain failure across a distribution region; and the compound loss of an input, energy, or water dependency that the cross-sector web propagates.

Section 5 — Dual Design Basis: Bright Line and Command Broker

[Skeleton — to be developed on the DBA-W §5.5 model.]

Agriculture and food production are rapidly automating — GPS-guided precision agriculture, automated feeding and milking, greenhouse and animal-facility climate control, food-processing control systems, and cold-chain monitoring. The Bright Line must be placed for any action of a component within scope of FD-LD §4.3 whose incorrect performance could compromise food safety or animal welfare at or above the severity threshold — an automated dosing or feed-formulation decision, a processing-control setpoint, a contamination-screening disposition — with the architecturally enforced authorization of a qualified operator or food-safety officer as Command Broker. The water sector's treatment-dosing analysis (DBA-W §5.5) is the nearest model. Determinism decides scope under FD-LD §4.3; it does not decide placement.

Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

6.1 Inward View — the Five-Sector Web

Food and Agriculture is the framework's most cross-sector-dependent sector, and its inward view is a web rather than a chain. It depends on energy for irrigation, processing, and refrigeration; on water for irrigation, livestock, and processing; on transportation for farm-to-market and distribution; on the chemical sector for fertilizers and pesticides; and on financial services for the commodity markets and crop insurance that underwrite production. Each of these is stated as a design basis condition, and the compound-failure scenario — in which more than one dependency degrades together — is the governing inward case because the web offers no single-dependency isolation.

6.2 Named Boundaries

Four boundaries are named for clarity. Fertilizer and pesticide production is Chemical, not Food and Agriculture — the ammonia supply that the chemical framework already ties by its United Nations identification number to both water treatment and agricultural fertilizer is the concrete instance. So is the storage and retail of fertilizer chemicals: agricultural chemical retailers holding ammonium nitrate or anhydrous ammonia are governed by DBA-CHEM-DI-FC3 (Distribution, Blending and Supply Terminals) on the author's decision of 5 October 2026, because their governing consequence is an explosion or release, and DBA-FA-DR-FC3 does not take them. Veterinary pharmaceutical manufacturing is Healthcare. The cold chain is shared with Transportation. And the commodity-market and crop-insurance dependency is a boundary with the financial sector — the first dependency any producer sector has carried on Financial Services, and a reminder that the financial sector's systemic consequence reaches into the food supply.

6.3 Outward View — Population Health and Nutrition

The outward consequence of the sector runs directly to the population: a supply disruption produces shortage and economic disruption, and a contamination produces public-health harm at population scale. Because food is a universal dependency, the sector's failure is felt across every other sector's workforce and population, and its safety consequence — like the water sector's — is a life-safety consequence rather than an economic one.

Section 7 — Military-Action Extension Applicability

[Skeleton.] The food supply is a recognized target of deliberate action — agroterrorism through contamination or the introduction of crop or livestock disease, and the attack on concentrated processing or input capacity. The contamination consequence is attractive to an adversary precisely because it is temporally delayed and attribution-resistant, characteristics that align with the persistence and denial features of military action. The sector-level presumption toward MA applicability is strong for

concentrated processing plants, major input and fertilizer facilities, and the disease-surveillance function. Application remains a per-facility determination under the mandatory two-factor test — geolocation and strategic value.

Section 8 — Facility Class Register and Inheritance Model

8.1 Inheritance Model

Facility-class instruments of every type in the DBA-FA series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework, and inherit the water sector's contamination-consequence structure on the safety axis. Each instrument adds the food-function-specific consequence chain on both the safety and the supply axes.

8.2 Facility Class Register

The following table is the current facility-class register for the DBA-FA series. DBA-FA-FP-FC2 has issued as a working draft and the other function classes are at Anticipated status; agricultural-input production is shown as governed by the chemical and healthcare series. Designators were ratified 2026-08-09 and are sub-indexed by commodity and perishability.

Designator	Function class (function-primary)	Governing characteristic	Status	Primary regulatory anchor
DBA-FA-AG-FC1	Agricultural production — crop production, livestock operations, dairy farms, aquaculture	geographically distributed; disease-outbreak vulnerability	Anticipated	USDA; state departments of agriculture
DBA-FA-FP-FC2	Food processing & manufacturing — meat & poultry processing, dairy, grain milling, beverage, packaged food	concentration chokepoint (few large processors)	Issued (working draft)	USDA FSIS; FDA (FSMA)
DBA-FA-DR-FC3	Storage, distribution & retail — grain elevators, cold storage, warehouses, distribution centers, grocery	cold-chain dependency; farm-to-table logistics	Anticipated	FDA; cold-chain boundary with DBA-TRAN
DBA-FA-OpsCo†	Food safety & biosecurity coordination (coordination class) — USDA/FDA inspection, FSMA, disease surveillance	contamination detection; outbreak response	Anticipated (architecture)	USDA / FDA; state; FSMA
(boundary)	Agricultural inputs — fertilizer & pesticide production and fertilizer-chemical storage and retail, seed, veterinary pharmaceuticals	fertilizer/ammonia → DBA-CHEM; veterinary pharma → DBA-HPH	Governed elsewhere	see those series (input production)

Designators were ratified 2026-08-09. Facility-class numbers are sequential across the series; OpsCo is a coordination class and carries no FC number, mirroring DBA-ES-CntlCo. Grounded in the CIKR_v16 sector map (Food and Agriculture Core Infrastructure and Dependencies).

Section 9 — Development Status and Open Items

This is the initial working draft of the DBA-FA sector framework, grounded in the CIKR_v16 sector map. It establishes the standalone-series placement, the dual safety-and-supply consequence, the distributed-web framing and its concentration chokepoints, the function-primary decomposition (agricultural production, food processing, storage and distribution, and the OpsCo food-safety-and-biosecurity class) with a commodity-and-perishability sub-index, the inheritance of the water sector's contamination-consequence structure, and the five-sector dependency web with its named Chemical, Healthcare, Transportation, and Financial boundaries. The design-basis envelope (Section 4), the dual-DB detail (Section 5), and the MA applicability section (Section 7) are stubs to be developed. Open items at this version: develop the dual supply-continuity and contamination consequence measures; and reconcile the agricultural-input boundary with DBA-CHEM and the commodity-market boundary with DBA-FIN. At v0.5, DBA-FA-FP-FC2 (*Food Processing and Manufacturing*) has issued as the series' first facility-class instrument, physical, written for all processing commodities at class level with beverage and packaged-food manufacture at the depth of the sub-index, and the three reference events it proposed are conferred at Section 9.1. It calibrates the dual consequence measure of Section 3.6 and develops the candidate bounding events of Section 4.1 for the processing chokepoint. It governs the processing plant's ammonia refrigeration and reads its release consequence through the chemical series' shared model (*Food Processing and Manufacturing* §5.3, §10.4), which refines the Chemical boundary of Section 6.2 for that equipment. What it notes to this framework is taken up at the framework's next conformance pass: the five parameters, the event categories it develops ahead of Section 4, the three bands of the commodity and perishability sub-index, its declared residuals, the consequence-basis conditionality check, and the question whether the sector should state a requirement for the substitution of first-band products such as infant formula (*Food Processing and Manufacturing* §10.10). Further open item: the three reference-event instruments of Section 9.1.

9.1 Conferred Reference Events

On the author's decision of 14 October 2026, this framework confers a reference-event designator on each of the three events its first facility class proposed. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), reference-event instruments are written and held by Project 11, DBA Reference Events; this framework keeps the conferral. None of the three instruments is yet written. The designators are numbered in the order the proposing class lists its events; the number records conferral, not issue. Each designator is sector-primary and single.

From *Food Processing and Manufacturing*, three events are conferred: DBA-FA-RE-1, the ransomware attack on JBS, May and June 2021; DBA-FA-RE-2, the shutdown of Abbott Nutrition's infant-formula plant at Sturgis, Michigan, and the shortage that followed, 2022; and DBA-FA-RE-3, the adulteration of milk with melamine in China, 2008. Highly pathogenic avian influenza, which the plan for closing the seven sectors named for this sector, is not conferred

here; it is held for *Agricultural Production* (DBA-FA-AG-FC1), to be proposed by that class when it is written. The three instruments are owed, and they are Project 11's to write. *Food Processing and Manufacturing* states the three as proposed at its v0.1 and conforms to the conferral at its next revision.

This framework is still a skeleton at Sections 4, 5 and 7. The three instruments read their events back through the design basis of *Food Processing and Manufacturing*, which develops it ahead of the framework, and what they return to the framework is taken up at its next conformance pass.

References

[To be completed.] Governing and inherited instruments: FD-BL and determination annexes; DBA-MA Parent Framework (IAEA L1); DBA Series Gaps and Recommendations (§1.7, MA applicability); DBA-W (contamination-consequence structure; §5.5 dosing analysis); DBA-CHEM (fertilizer/pesticide and ammonia boundary, UN identification link); DBA-TRAN (cold-chain boundary); DBA-HPH (veterinary pharmaceutical boundary); DBA-FIN (commodity-market and crop-insurance dependency); DBA-EN (irrigation, processing, refrigeration power). Regulatory: USDA and HHS (co-SSA); USDA FSIS; FDA (FSMA); EPA; state departments of agriculture. Grounding: CIKR_v16 sector map (Food and Agriculture Sector). Series instruments: DBA-FA-FP-FC2 (*Food Processing and Manufacturing*), the series' first facility class; DBA-FA-RE-1 to DBA-FA-RE-3, the conferred reference events of Section 9.1, their instruments owed to Project 11. DBA-CHEM-PR-FC1 (*Chemical Production and Synthesis*), whose shared release model the class cites for ammonia refrigeration.

Revision History

0.5 October 2026 — T. Roxey. Section 9.1 is added and confers DBA-FA-RE-1 to DBA-FA-RE-3 on the three reference events that DBA-FA-FP-FC2 (Food Processing and Manufacturing) proposed, on the author's decision of 14 October 2026: the JBS ransomware attack (May–June 2021), the Abbott Sturgis shutdown and infant-formula shortage (2022) and the melamine adulteration of milk in China (2008). Highly pathogenic avian influenza is held for Agricultural Production. Each designator commits the series to a register row and a standalone instrument, none yet written; the instruments are Project 11's. The register at Section 8.2 enters the class as issued. Section 3.6 records the class's calibration of the dual consequence measure. Section 9 records the class's issue, its ammonia-refrigeration boundary with the chemical series, and what it notes to this framework for the next conformance pass. References add the class, the three designators and DBA-CHEM-PR-FC1. No change to the taxonomy, the designators of the classes, the sub-index or the design basis. Revision-history entries are left as records.

0.4 October 2026 — T. Roxey. The boundary with the Chemical sector at Section 6 and in the register now includes the storage and retail of fertilizer chemicals: agricultural chemical retailers holding ammonium nitrate or anhydrous ammonia are governed by DBA-CHEM-DI-FC3 on the author's decision of 5 October 2026, and DBA-FA-DR-FC3 does not take them. Revision-

history entries are left as records. No change to the taxonomy, the designators or the design basis.

0.3 October 2026 — T. Roxey. Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026. Section 5 states the scope of Bright Line placement by the determinism carve-out of FD-LD §4.3 rather than as any AI action, and states that determinism decides scope and not placement (P9-11, wording). Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its sense: the decomposition tier reads function class, and the coordination layer reads coordination class (OI-15). No change to the design basis, the taxonomy or the register.

0.2 August 2026 — T. Roxey. Designators ratified; codes PA → AG and DS → DR (collision/clarity fixes); coordination resolved to OpsCo (producer under continuous food-safety regulation); dual supply+safety consequence confirmed.

0.1 August 2026 — T. Roxey. Initial skeleton, grounded in the CIKR_v16 sector map. Established DBA-FA as a standalone sector series with a dual safety-and-supply consequence (inheriting the water sector's contamination-consequence structure on the safety axis and adding the crop/livestock biosecurity dimension) and the distributed-web framing (production distributed with concentration chokepoints at processing and inputs; the most cross-sector-coupled sector, dependent on energy, water, transportation, chemical, and financial services). Adopted the function-primary decomposition (Agricultural Production / Food Processing / Storage, Distribution & Retail / OpsCo food-safety-and-biosecurity) with a commodity-and-perishability sub-index; named the Chemical (fertilizer/ammonia), Healthcare (veterinary pharma), Transportation (cold chain), and Financial (commodity markets/crop insurance) boundaries. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.